Chapter 6

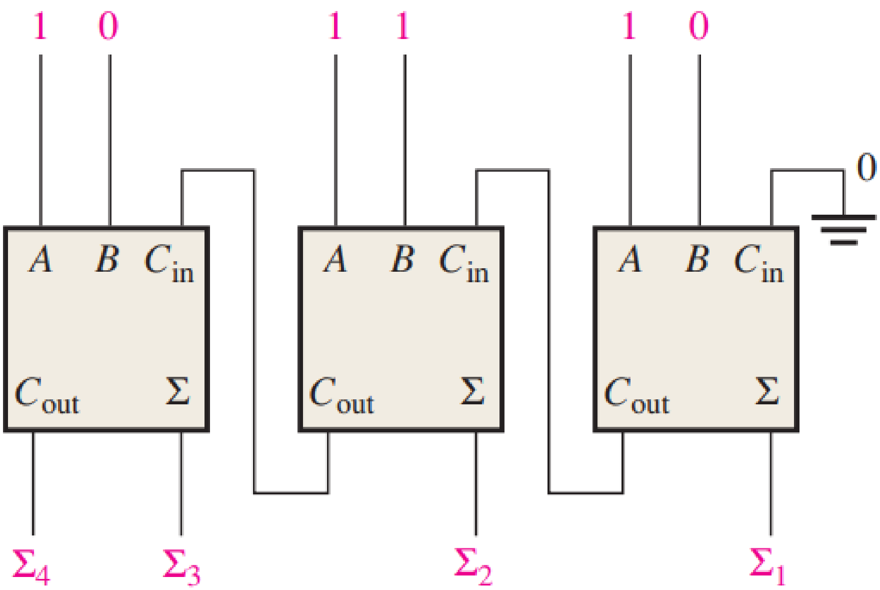
2024.2.22

Name:Chen Zhuoyan Student ID:523030910037

2.What are the full-adder inputs that will produce each of the following outputs.
(a)$\Sigma = 0, C_{out} = 0$ (b) $\Sigma = 1, C_{out} = 0$
(c)$\Sigma = 1, C_{out} = 1$ (d)$\Sigma = 0, C_{out} = 1$

Answer Area:
(a)$A = 0, B = 0, C_{in} = 0$
(b)$A = 0, B = 0, C_{in} = 1$, or
$A = 0, B = 1, C_{in} = 0$, or
$A = 1, B = 0, C_{in} = 0$, or
(c)$A = 1, B = 1, C_{in} = 1$
(d)$A = 1, B = 1, C_{in} = 0$, or
$A = 1, B = 0, C_{in} = 1$, or
$A = 0, B = 1, C_{in} = 1$

4.For the parallel adder in Figure 6-57, determine the complete sum by analysis of the logical operation of the circuit. Verify your result by longhand addition of the two inputs numbers.

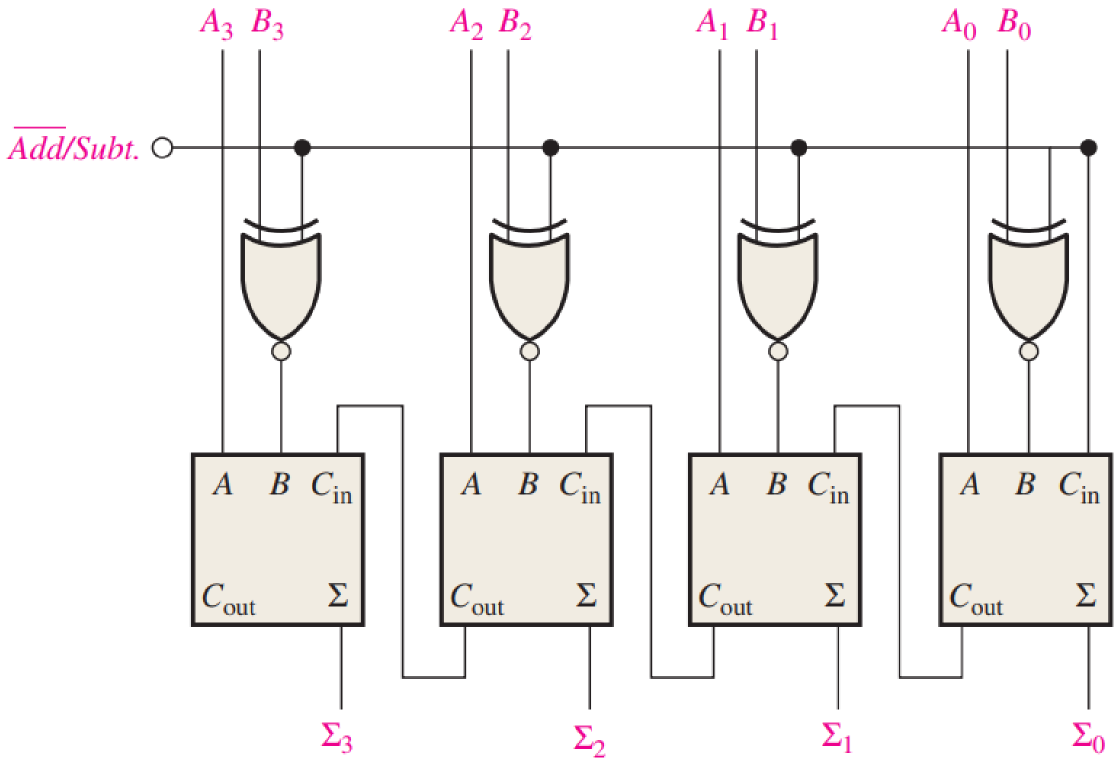


Answer Area:
For the first adder: $\Sigma_1 = \Sigma = 0, C_{out} = 1$
For the second adder: $C_{in} = 1, \Sigma_2 = \Sigma = 0, C_{out} = 1$
For the third adder: $C_{in} = 1, \Sigma_3 = \Sigma = 1, \Sigma_4 = C_{out} = 1$
Therefore, the complete sum is 1100. According to the longhand addition:

$$\begin{array}{rr} & 111 \\ + & 101 \\ \hline & 1100 \end{array}$$

The sum is 1100.

6.The circuit shown in Figure 6–71 is a 4-bit circuit that can add or subtract numbers in a form used in computers (positive numbers in true form; negative numbers in complement form). (a) Explain what happens when the $\overline{Add}/Subt.$ input is HIGH. (b) What happens when $\overline{Add}/Subt.$ is LOW?



Answer Area:
(a)When the the $\overline{Add}/Subt.$ input is HIGH.Then for the first Adder, $C_{in} = 1$
$\because B_i\overline{Add}/Subt = \overline{B_i}$
$\therefore$ The result is equal to $A_3A_2A_1A_0 - B_3B_2B_1B_0$
(2))When the the $\overline{Add}/Subt.$ input is LOW.Then for the first Adder, $C_{in} = 0$
$\because B_i\overline{Add}/Subt = B_i$
$\therefore$ The result is equal to $A_3A_2A_1A_0 + B_3B_2B_1B_0$

7.For the circuit in Figure 6–71, assume the inputs are $\overline{Add}/Subt. = 1$, A = 1001, and B = 1101. What is the output?
Answer Area:
According to Problem 6, the output is A - B
$A = 1001 = -7 \quad B = 1100 = -4$
Then A - B = 1001 + 0011 + 1 = 1101

10.In the process of checking a 74HC283 4-bit parallel adder, the following logic levels are observed on its pins: 1-HIGH, 2-HIGH, 3-HIGH, 4-HIGH, 5-LOW, 6-LOW, 7-LOW, 9-HIGH, 10-LOW, 11-HIGH, 12-LOW, 13-HIGH, 14-HIGH, and 15-HIGH. Determine if the IC is functioning properly.
Answer area:
The adder is calculating 110 + 1110 = 10100, Which can't match the output 111.

Therefore, the IC is not functioning.

11.Each of the eight full-adders in an 8-bit parallel ripple carry adder exhibits the following propagation delay:

$A to \Sigma and C_{out}$: $40ns$

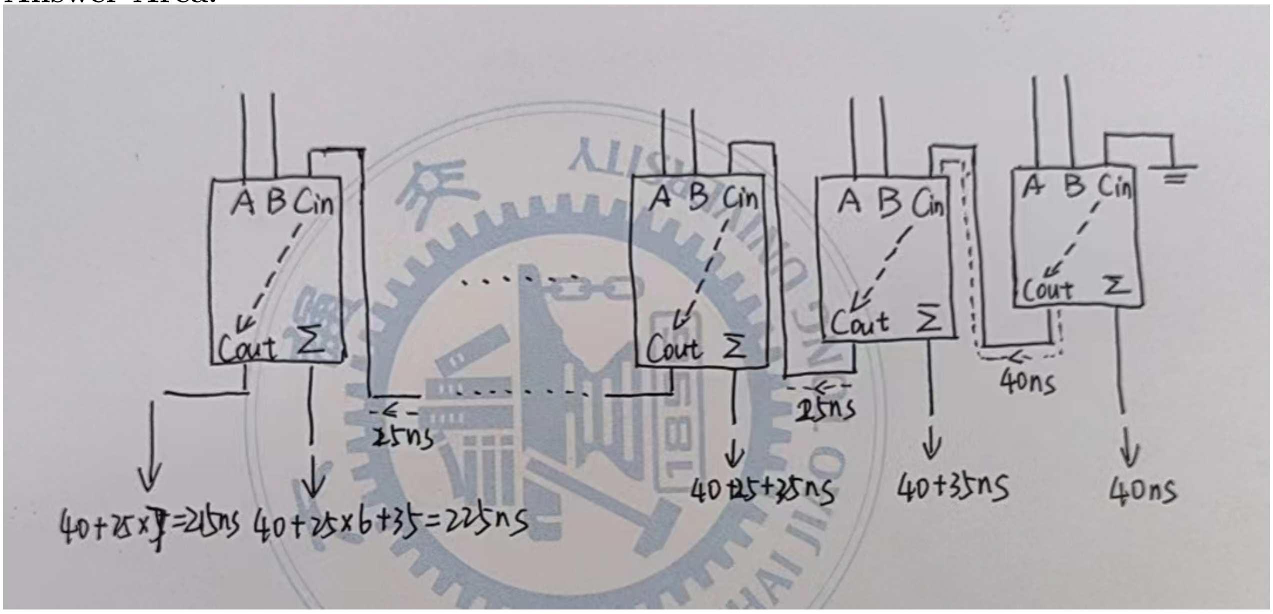
$B to \Sigma and C_{out}$: $40ns$

$C_{in} to \Sigma$: $35ns$

$C_{in} to C_{out}$: $25ns$

Determine the maximum total time for the addition of two 8-bit numbers.

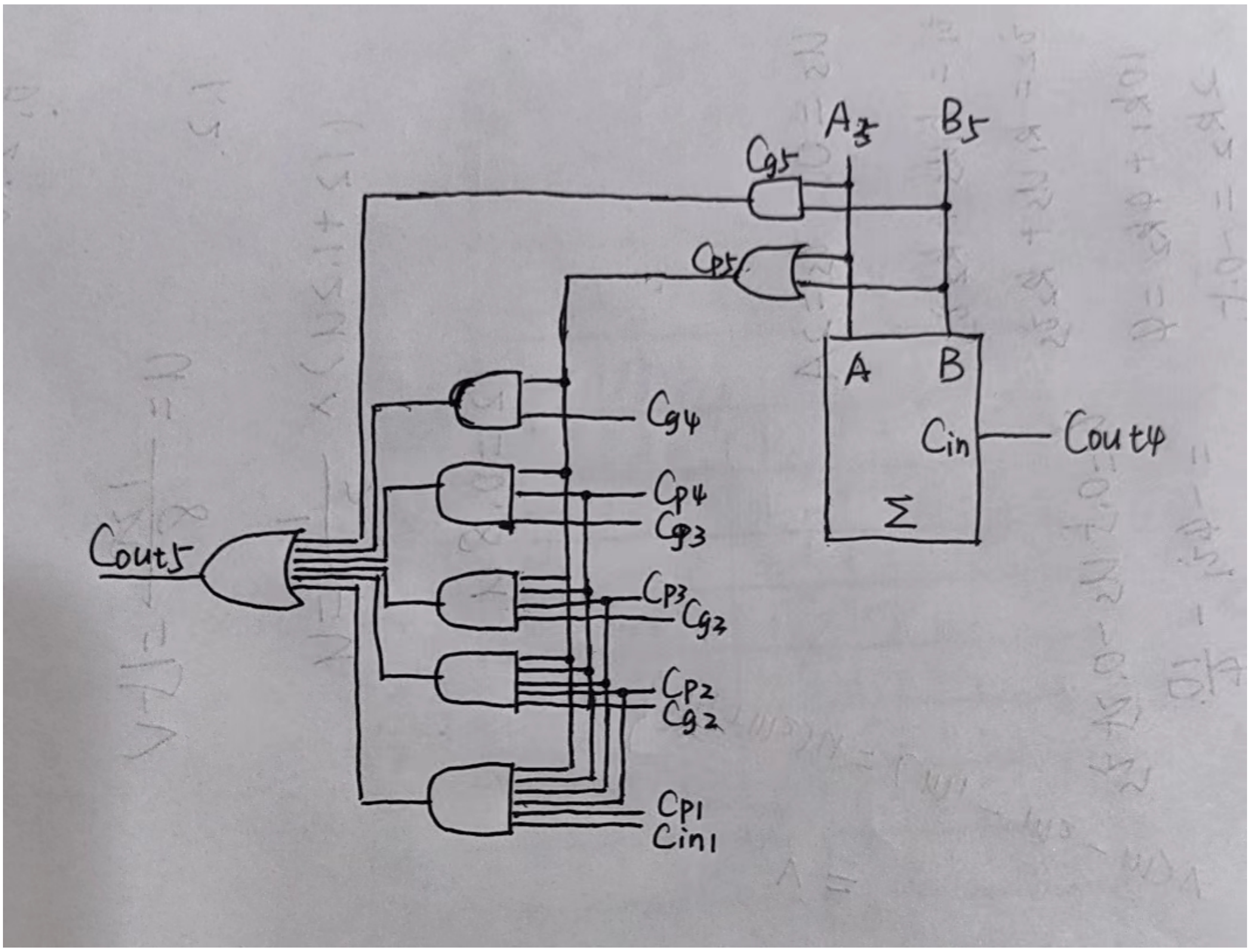
Answer Area:



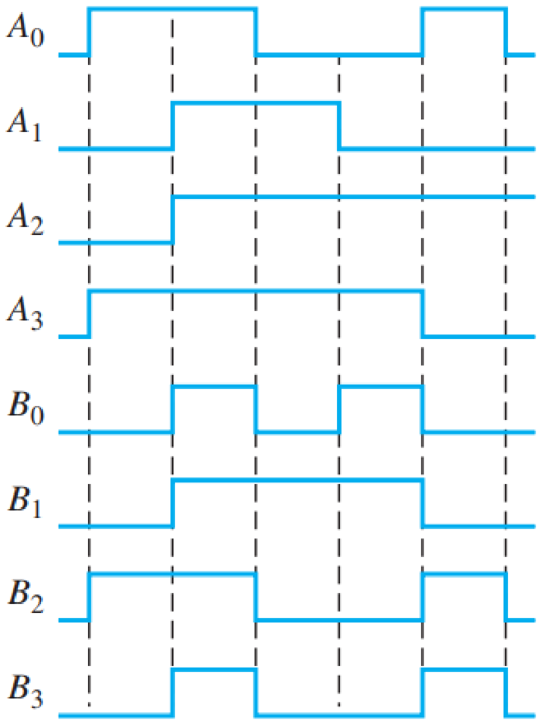
$t = 40 + 6 \times 25 + 35 = 225ns$

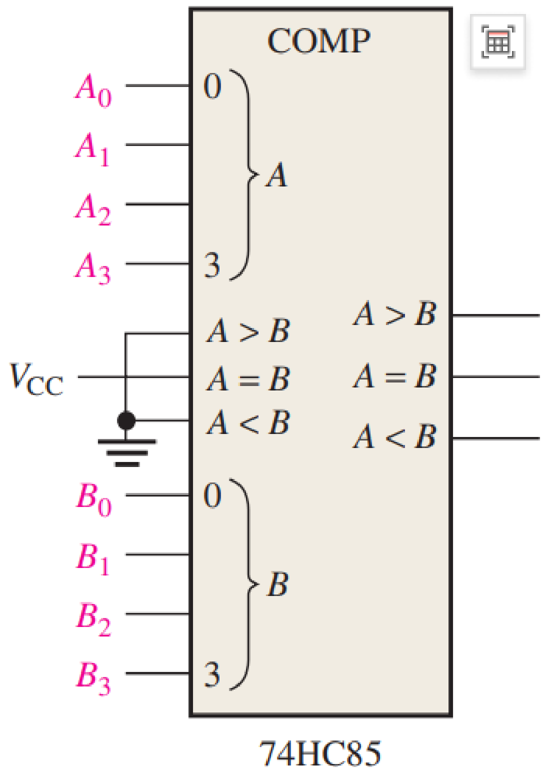
12.Show the additional logic circuitry necessary to make the 4-bit look-ahead carry adder in Figure 6–17 into a 5-bit adder.

Answer Area:

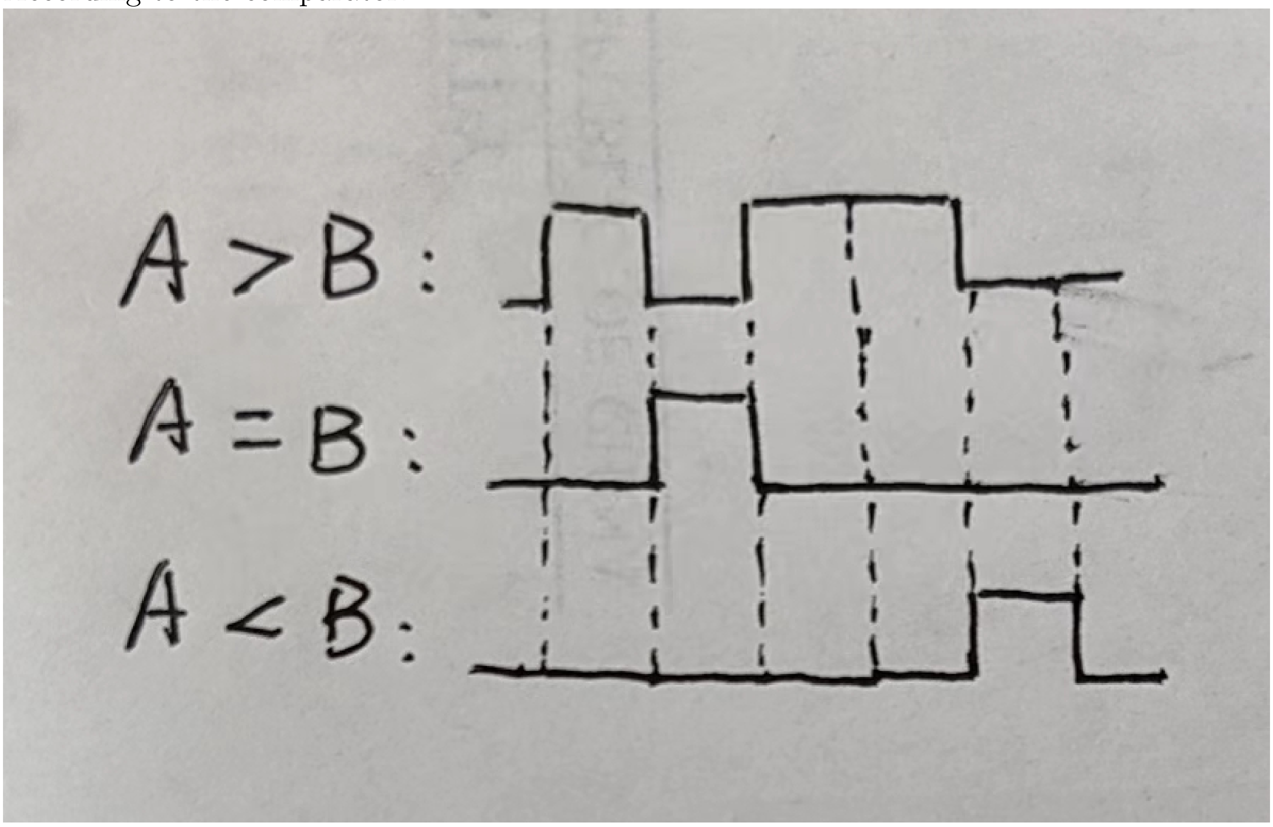


14.For the 4-bit comparator in Figure 6–74, plot each output waveform for the inputs shown. The outputs are active-HIGH.





According to the comparator:

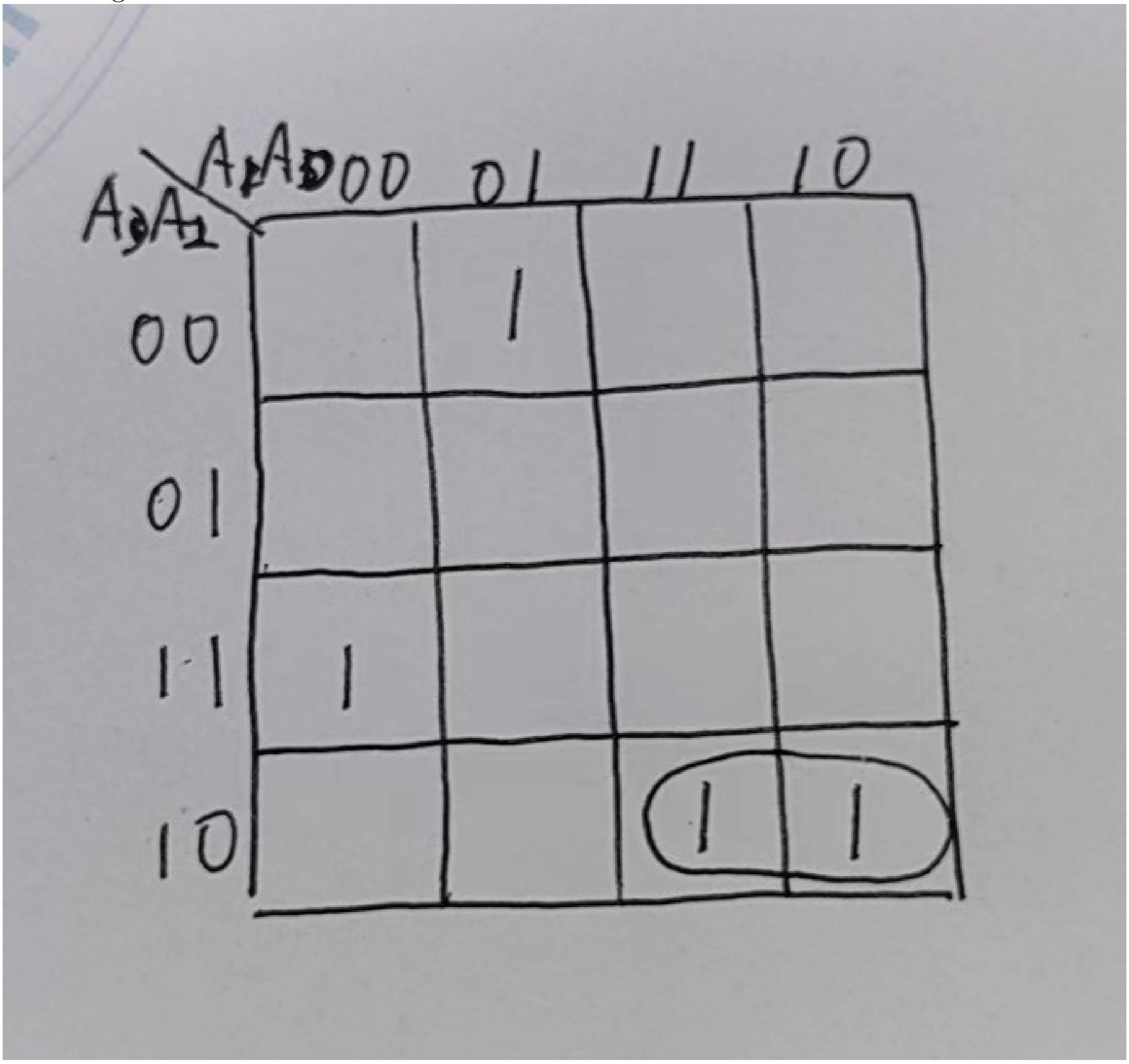


19.You wish to detect only the presence of the codes 1010, 1100, 0001, and 1011. An active-HIGH output is required to indicate their presence. Develop the minimum decoding logic with a single output that will indicate when any one of these codes is on the inputs. For any other code, the output must be LOW.:

Answer Area:

$X = A_3\overline{A_2}A_1\overline{A_0} + A_3A_2\overline{A_1A_0} + \overline{A_3A_2A_1}A_0 + A_3\overline{A_2}A_1A_0$

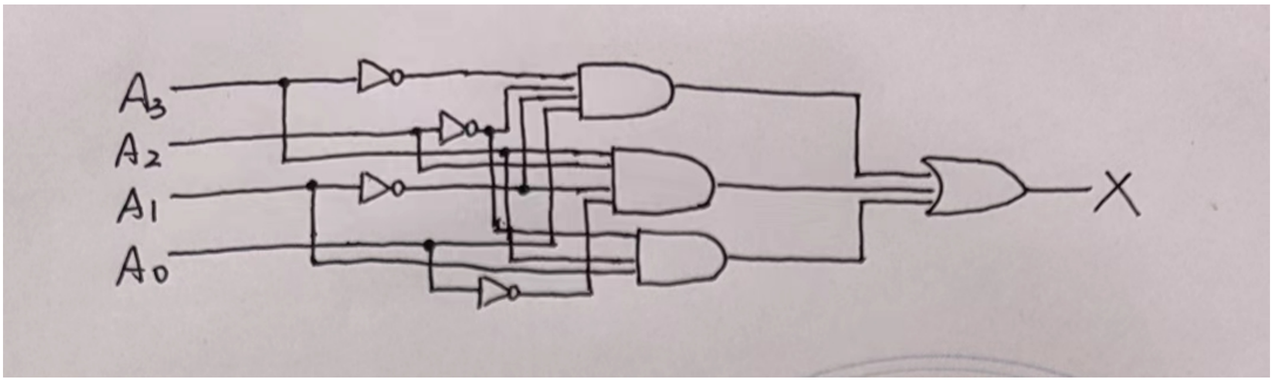
According to:



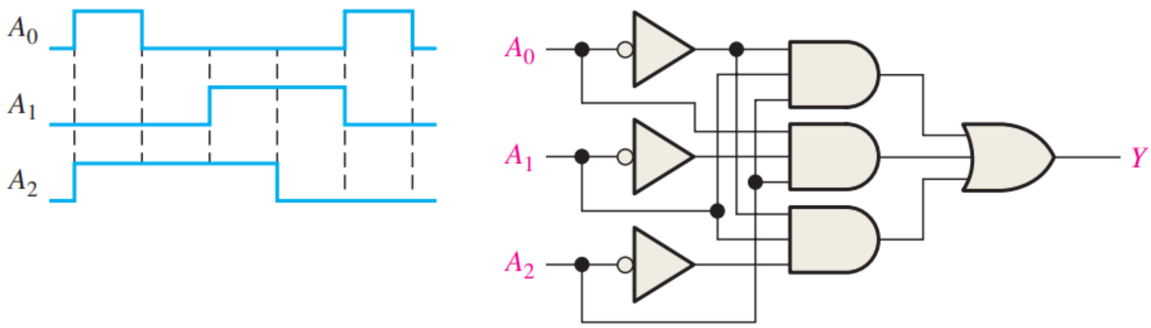
Then:

$X = A_3A_2\overline{A_1}\cdot\overline{A_0} + \overline{A_3}\cdot\overline{A_2}\cdot\overline{A_1}A_0 + A_3\overline{A_2}A_1$

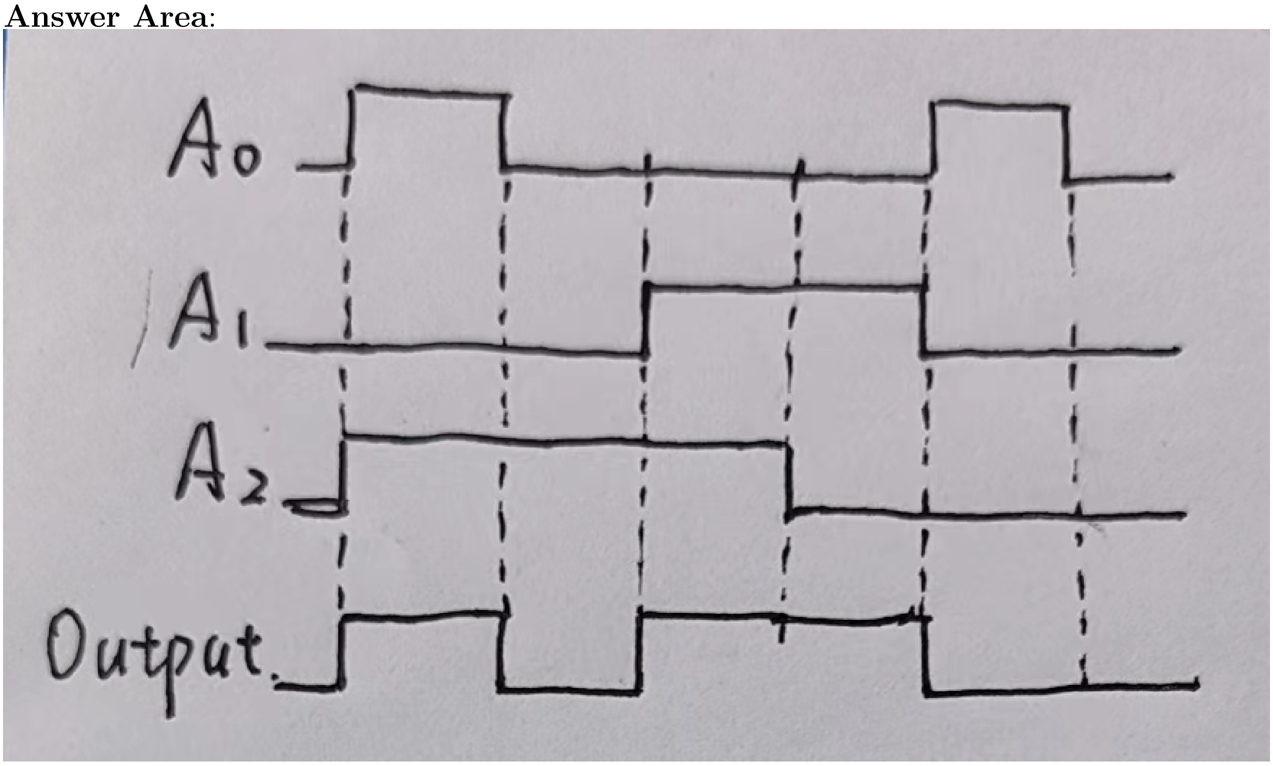
Then:



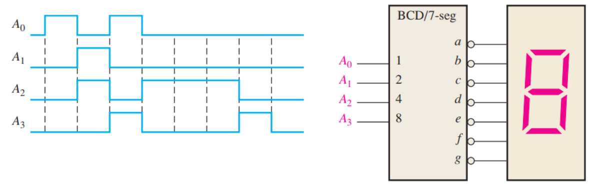
20.If the input waveforms are applied to the decoding logic as indicated in Figure 6–76, sketch the output waveform in proper relation to the inputs.



Answer Area:



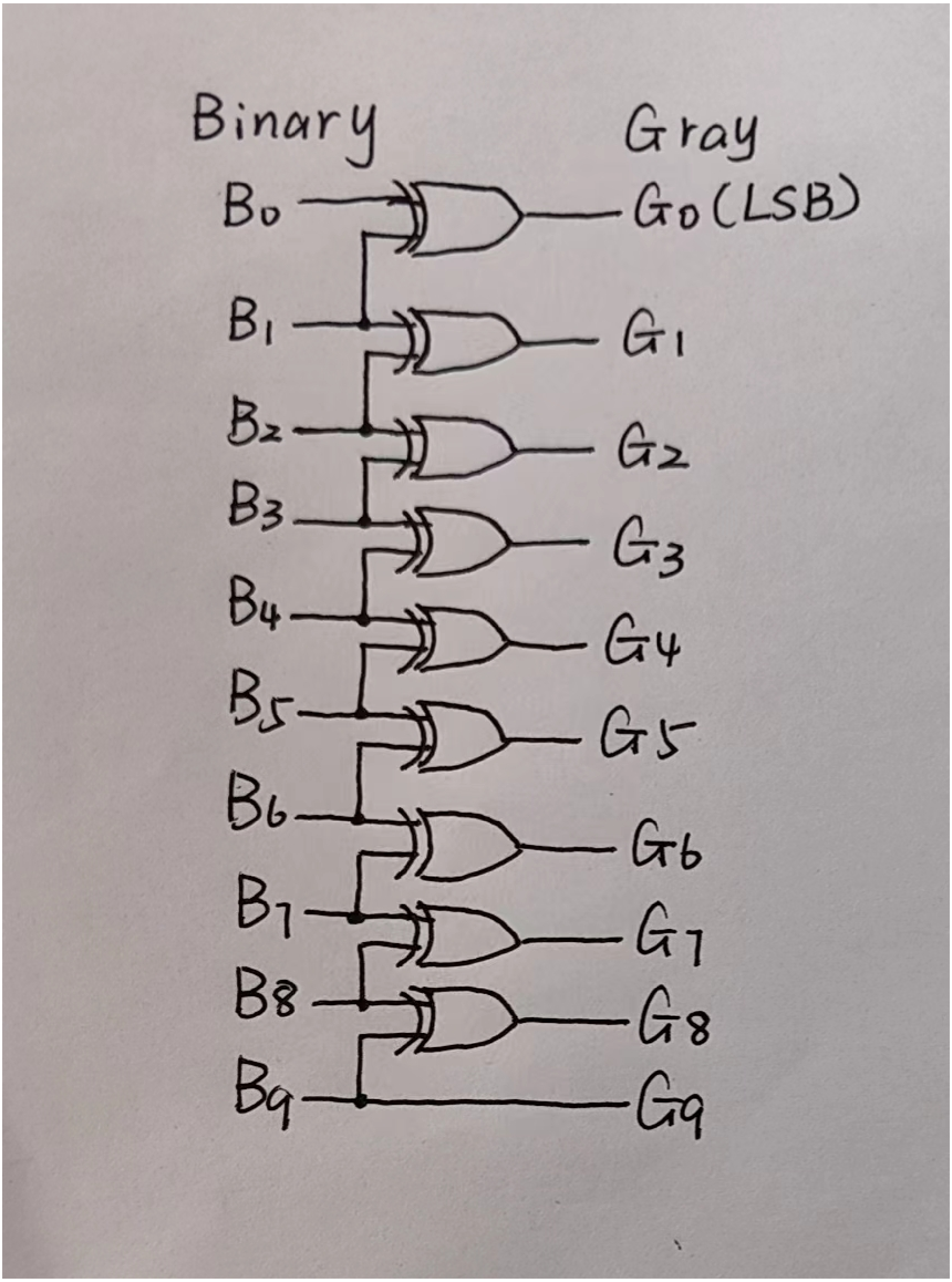
22.A 7-segment decoder/driver drives the display in Figure 6–78. If the waveforms are applied as indicated, determine the sequence of digits that appears on the display.



Answer Area:
The sequence of inputs is 0001, 0110, 1001, 0100, 0100, 0100, 1000, 0000.
Then the sequence of digits that appears on the display is 1, 6, 9, 4, 4, 4, 8, 0.

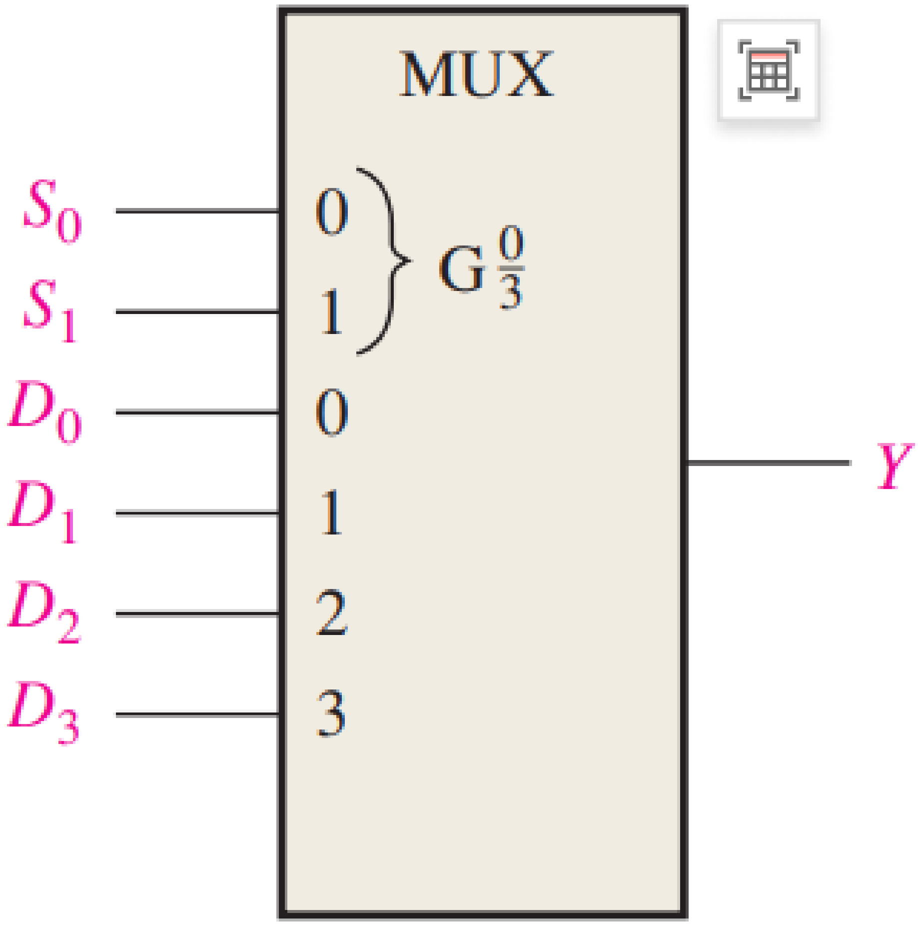
24.A 74HC147 encoder has LOW levels on pins 2, 5, and 12. What BCD code appears on the outputs if all the other inputs are HIGH?
Answer Area:
$\because$ The encoder will produce a BCD output corresponding to the highest-order decimal digit input that is active and will ignore any other lower-order active inputs.
$\therefore$ The BCD code appears on the outputs is 1000.

26.Show the logic required to convert a 10-bit binary number to Gray code and use that logic to convert the following binary numbers to Gray code.
(a) 1010101010 (b) 1111100000 (c) 0000001110 (d) 1111111111
Answer Area:



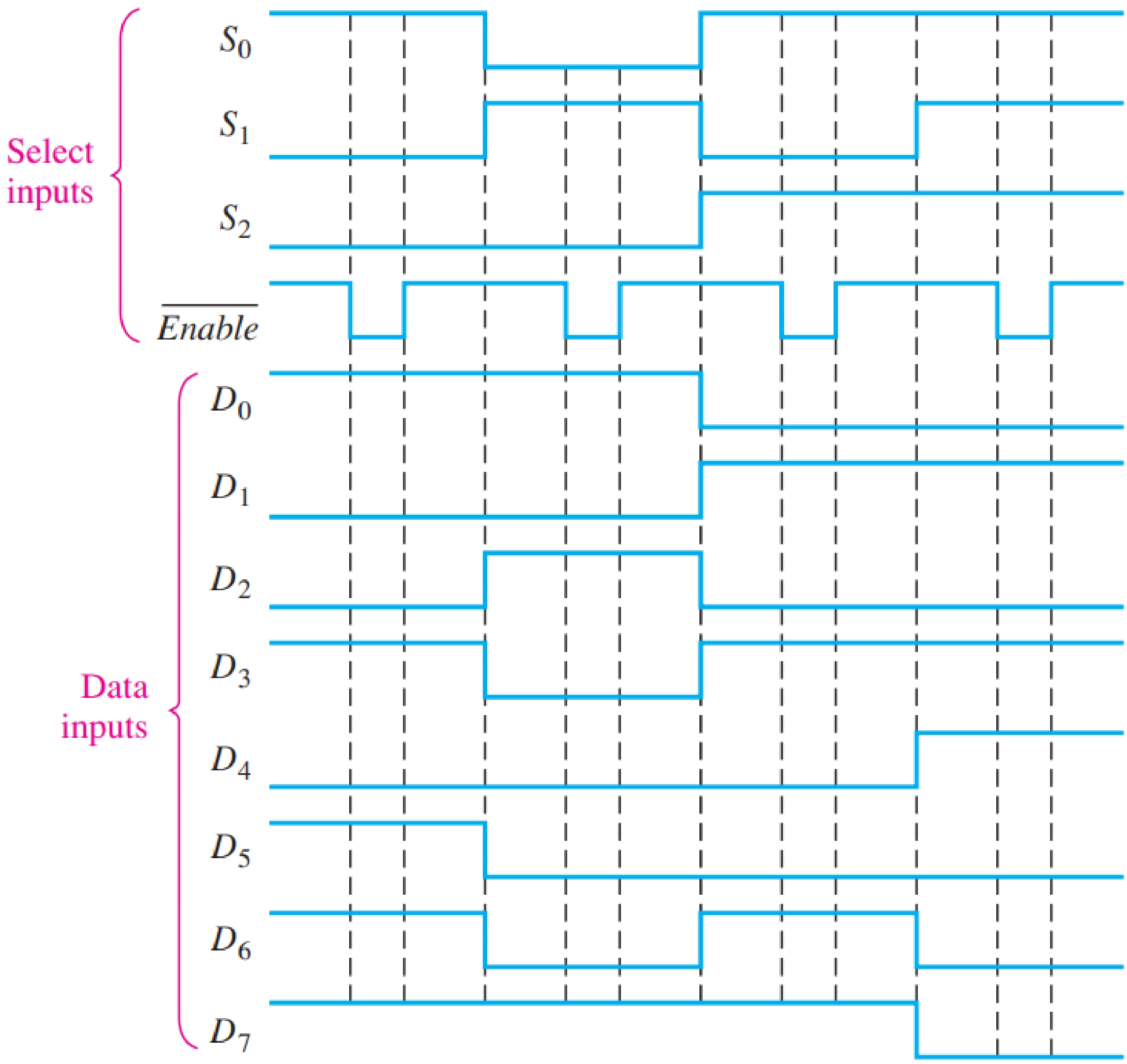
(a) Binary : 1010101010; Gray : 1111111111
(b) Binary : 1111100000; Gray : 1000010000
(c) Binary : 0000001110; Gray : 0000001001
(d) Binary : 1111111111; Gray: 1000000000

28.For the multiplexer in Figure 6–79, determine the output for the following input states: $D_0 = 0, D_1 = 1, D_2 = 1, D_3 = 0, S_0 = 1, S_1 = 0$.



Answer Area:
(c)
$\because S_1S_0 = 01$
$\therefore Y = D_1 = 1$

30.The waveforms in Figure 6–81 are observed on the inputs of a 74HC151 8-input multiplexer. Sketch the Y output waveform.



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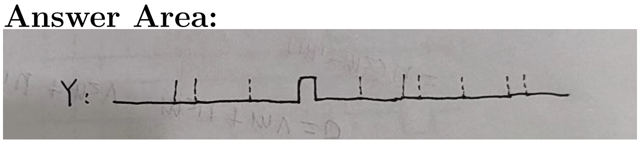
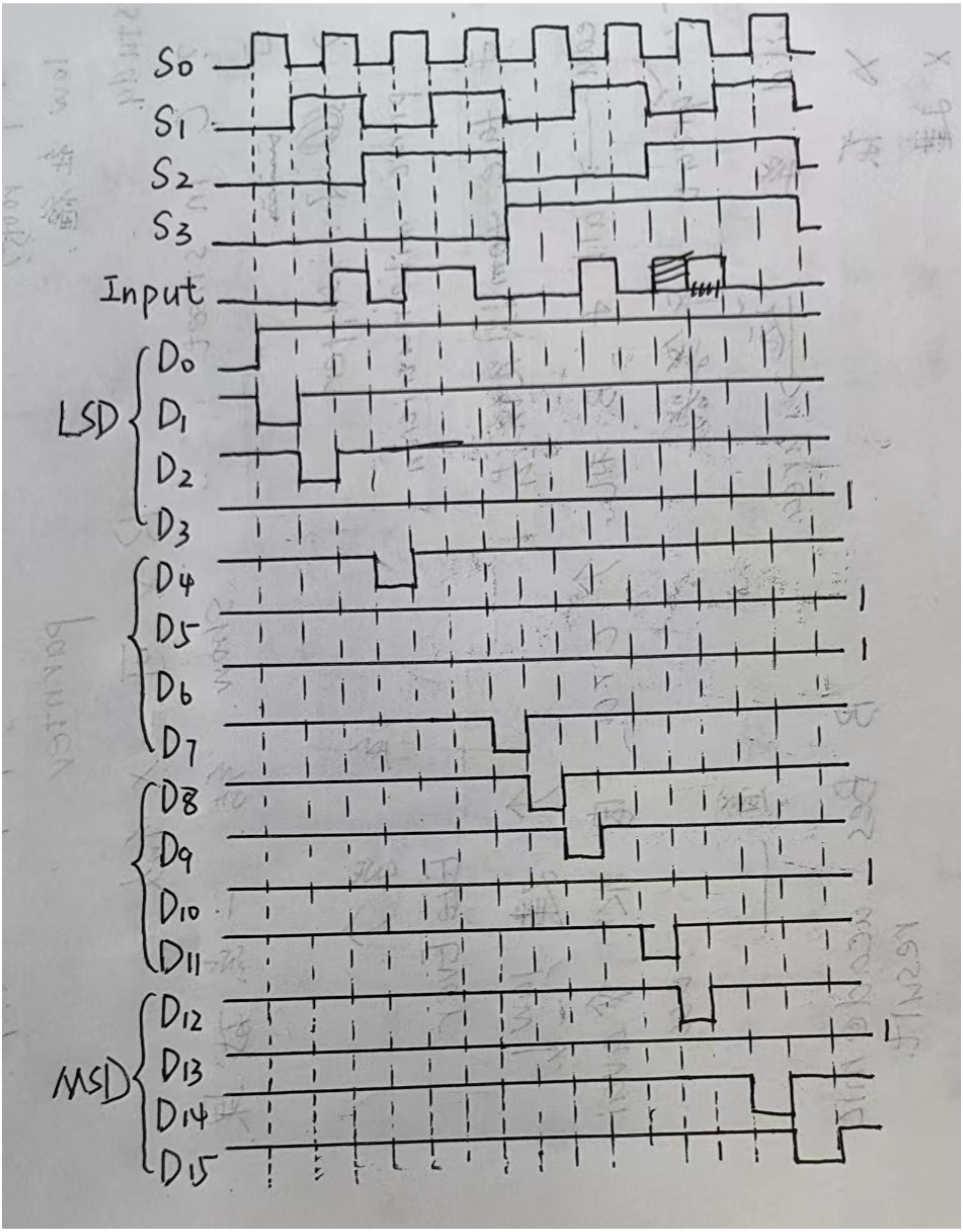
Answer Area:

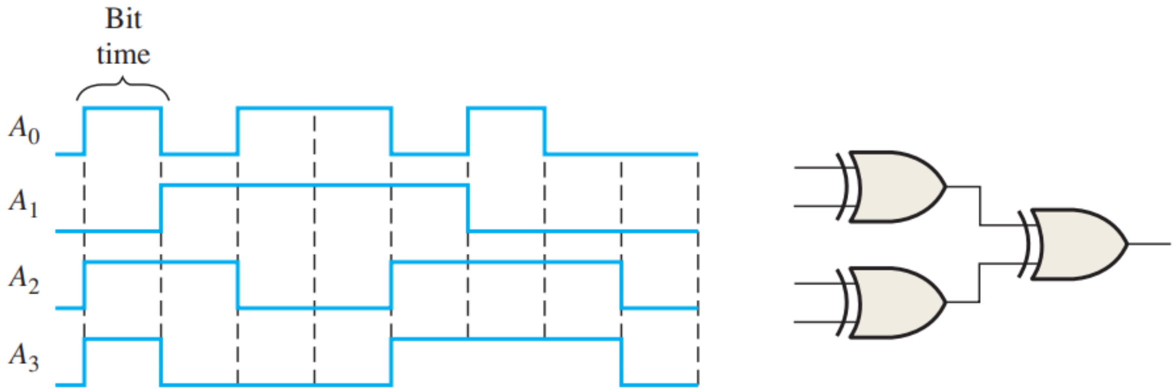


31.Develop the total timing diagram (inputs and outputs) for a 74HC154 used in a demultiplexing application in which the inputs are as follows: The data-select inputs are repetitively sequenced through a straight binary count beginning with 0000, and the data input is a serial data stream carrying BCD data representing the decimal number 2468. The least significant digit (8) is first in the sequence, with its LSB first, and it should appear in the first 4-bit positions of the output.

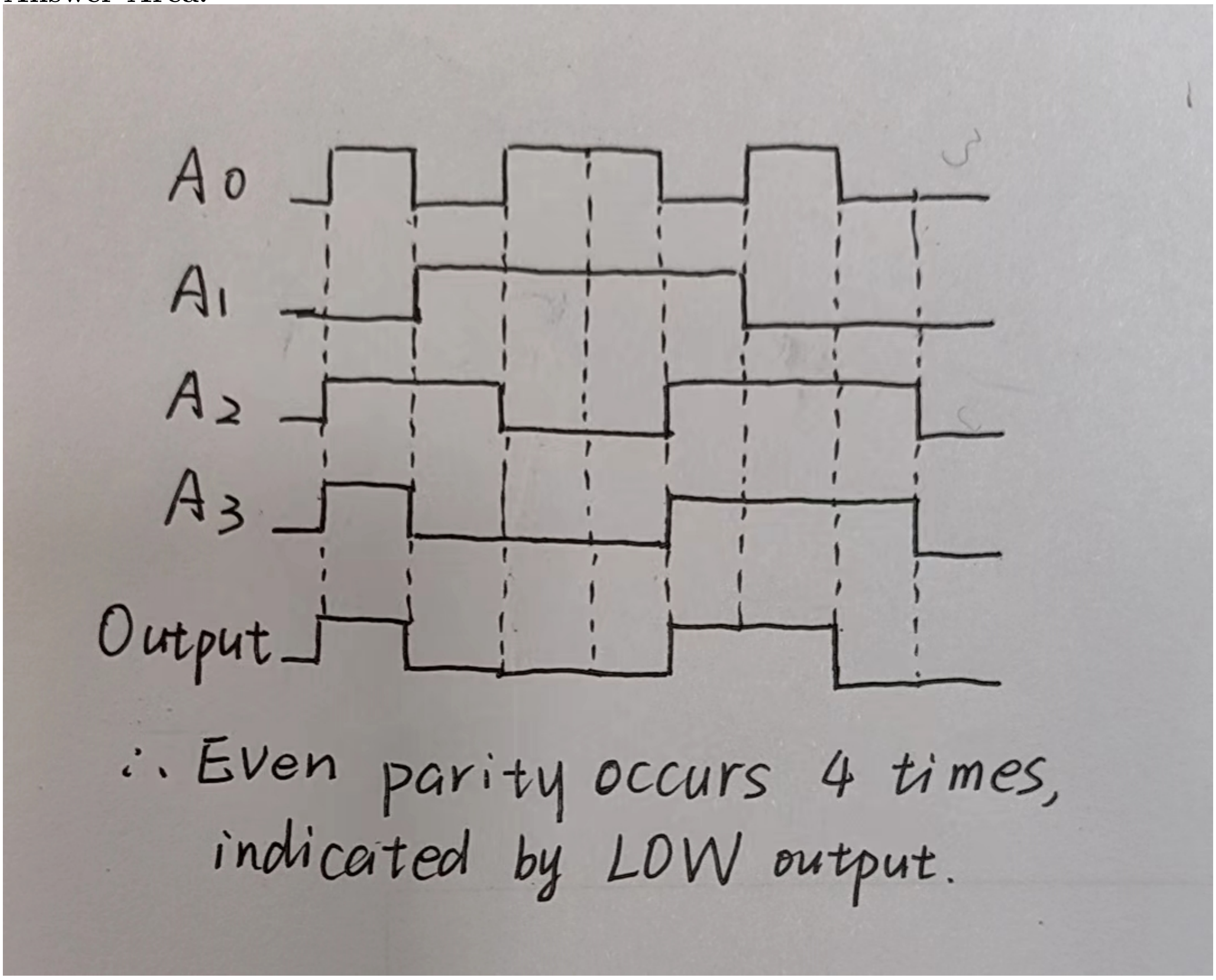
Answer Area:



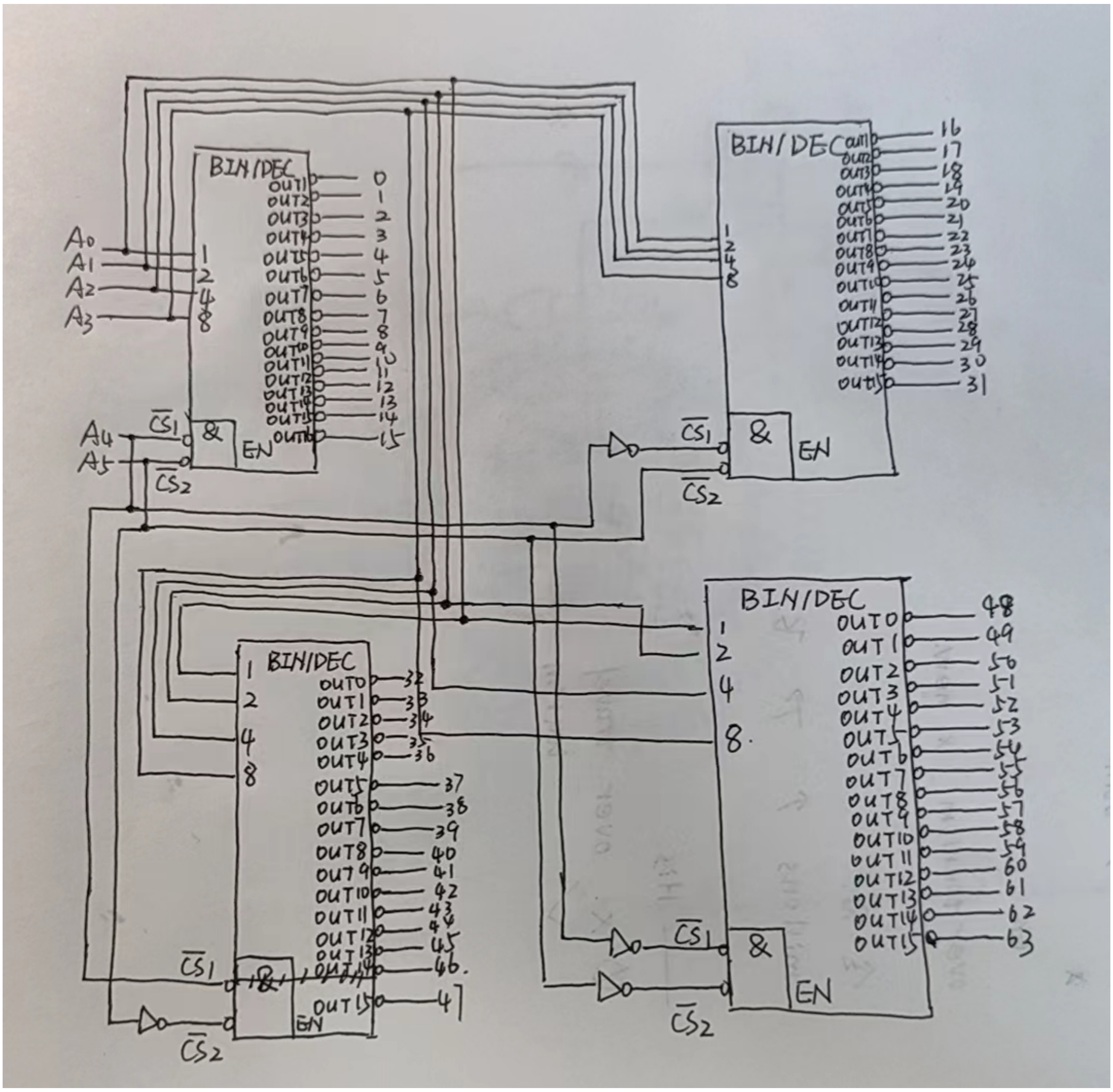
32.The waveforms in Figure 6–82 are applied to the 4-bit parity logic. Determine the output waveform in proper relation to the inputs. For how many bit times does even parity occur, and how is it indicated? The timing diagram includes eight bit times.



Answer Area:



6-bit decoder



74LS147

Truth Table:

$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$	$\overline{8}$	$\overline{9}$	$\overline{A_0}$	$\overline{A_1}$	$\overline{A_2}$	$\overline{A_3}$
0	0	0	0	0	0	0	0	0	0	0	0	0
×	×	×	×	×	×	×	×	1	1	0	0	1
×	×	×	×	×	×	×	1	0	0	0	0	1
×	×	×	×	×	×	1	0	0	1	1	1	0
×	×	×	×	×	1	0	0	0	0	1	1	0
×	×	×	×	1	0	0	0	0	1	0	1	0
×	×	×	1	0	0	0	0	0	0	0	1	0
×	×	1	0	0	0	0	0	0	1	1	0	0
×	1	0	0	0	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	1	0	0	0

Therefore

$\overline{A_0} = \overline{9} + 98\overline{7} + 9876\overline{5} + 987654\overline{3} + 98765432\overline{1} = \overline{9} + 8(\overline{7} + 6(\overline{5} + 4(\overline{3} + 2\overline{1}))) = \overline{9} + 8\overline{7} + 86\overline{5} +$

$864\overline{3} + 8642\overline{1}$
$\overline{A_1} = 98\overline{7} + 987\overline{6} + 987654\overline{3} + 9876543\overline{2} = 98\overline{7} + 98\overline{6} + 9854\overline{3} + 9854\overline{2}$
$\overline{A_2} = 98\overline{7} + 987\overline{6} + 9876\overline{5} + 98765\overline{4} = 98(\overline{7} + \overline{6} + \overline{5} + \overline{4})$
$\overline{A_3} = \overline{9} + 9\overline{8} = \overline{9} + \overline{8}$
Therefore
$A_0 = 9(\overline{8} + 7)(\overline{8} + \overline{6} + 5)(\overline{8} + \overline{6} + \overline{4} + 3)(\overline{8} + \overline{6} + \overline{4} + \overline{2} + 1)$
$A_1 = (\overline{9} + \overline{8} + 7)(\overline{9} + \overline{8} + 6)(\overline{9} + \overline{8} + \overline{5} + \overline{4} + 3)(\overline{9} + \overline{8} + \overline{5} + \overline{4} + 2)$
$A_2 = \overline{9} + \overline{8} + 7654$
$A_3 = 98$